L24

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Power and Spectral Signals:

Let $f(t)$ be a function of time. The autocorrelation matrix is defined as:-

$$R_{uu}(\tau) = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} f(t+\tau) f^*(t)\, dt$$

$$\boxed{R_{uu}(\tau) = R^*_{uu}(-\tau)}$$

if the limit exists and is finite for all τ.

If the F.T. of $R_{uu}(\tau)$ exists, then

$$S_{uu}(j\omega) = \int_{-\infty}^{\infty} R_{uu}(\tau) e^{-j\omega\tau} d\tau$$ is the spectral density function.

Power signal: (I) $R_{uu}(\tau)$ exists and is finite for all τ.

(II) $S_{uu}(j\omega)$ exists (note that $S_{uu}(j\omega)$ may not be bounded, impulses are allowed)

The power-norm of the signal $f(t)$ is given by

$$\left.\|f(t)\|_p = \sqrt{\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \|f(t)\|^2 dt} = \sqrt{\text{Trace}\left(R_{uu}(0)\right)}\right\} \text{semi-norm}$$

e.g. $f(t) = e^{-t} u(t)$

$$\|f(t)\|_p = \sqrt{\lim_{T\to\infty} \frac{1}{2T} \int_{0}^{T} e^{-t} dt} = \sqrt{\lim_{T\to\infty} \frac{1}{2T}\left[1 - e^{-T}\right]} = 0$$

Set of power-signals: P $\longleftarrow$ Space

Power-norm of a signal: $\|f\|_p^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{Trace}\left[S_{uu}(j\omega)\right] d\omega$

Bounded power spectral density signals (Spectral signals):

$$S := \left\{ f(t) \in \mathbb{R}^m / \ \|S_{uu}(j\omega)\|_\infty < \infty \right\}$$

Spectral-density norm $\|f(t)\|_S := \sqrt{\|S_{uu}(j\omega)\|}$

Banach space:

		Norm
	$\ell_p[0,\infty)$	$\|x\|_p := \left(\sum_{i=0}^{\infty} \|x_i\|^p\right)^{1/p}$
	$\ell_\infty[0,\infty)$	$\|x\|_\infty := \sup_i \|x_i\|$
$1 \leq p < \infty$	$\mathcal{L}_p(\mathcal{I})$	$\|x\|_p = \left(\int_{\mathcal{I}} \|x(t)\|^p dt\right)^{1/p}$
	$\mathcal{L}_\infty(\mathcal{I})$	$\|x\|_\infty = \operatorname{ess\,sup}_{\mathcal{I}} \|x(t)\|$
	$\mathcal{L}_\infty(j\mathbb{R})$	$\|F\|_\infty := \operatorname{ess\,sup}_{\omega \in \mathbb{R}} \sigma_{max} \|F(j\omega)\|$

$\mathcal{RL}_\infty$ – real, rational, proper T.M with no poles on $j\mathbb{R}$.

$\mathcal{H}_\infty$ (analytic in $\mathbb{C}^+$ and bounded on $j\mathbb{R}$) $\|F\|_{\mathcal{H}_\infty} := \operatorname{ess\,sup}_{\omega \in \mathbb{R}} \sigma_{max} |F(j\omega)|$

$\mathcal{RH}_\infty$ – real, rational, proper T.M with no poles in $\bar{\mathbb{C}}^+$.

$\mathcal{H}_\infty^-$ (analytic in $\mathbb{C}^-$ and bounded on $j\mathbb{R}$ $\|F\|_{\mathcal{H}_\infty} := \operatorname{ess\,sup}_{\omega \in \mathbb{R}} \sigma_{max} |F(j\omega)|$

$\mathcal{RH}_\infty^-$ – real, rational, proper T.M with antistable poles.

Power signal: $\mathcal{P}$ $\quad \|x\|_{\mathcal{P}} = \sqrt{\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \|x\|_2^2 \, dt}$

Spectral signals: $\mathcal{S}$ $\quad \|x\|_{\mathcal{S}} = \sqrt{\|S_{xx}(j\omega)\|_\infty}$

Hilbert space:

$\ell_2(-\infty, \infty)$ $\qquad \|x\|_2 := \sqrt{\sum_{i=0}^{\infty} |x_i|^2}$ $\qquad \langle x, y\rangle = \sum_{i=-\infty}^{\infty} x_i^* y_i$

$\mathcal{L}_2(\mathcal{I})$ $\qquad \|x\|_2 := \sqrt{\int_{\mathcal{I}} x^* x \, dt}$

e.g. $\mathcal{L}_2(-\infty, \infty)$, $\mathcal{L}_2[0, \infty)$, $\mathcal{L}_2(-\infty, 0)$

$\mathcal{L}_2 = \mathcal{L}_{2+} \oplus \mathcal{L}_{2-}$

$\mathcal{L}_2(j\mathbb{R})$ $\qquad \|F\| := \sqrt{\frac{1}{2\pi} \int_{-\infty}^{\infty} \text{trace}\left(F^*(j\omega) F(j\omega)\right) d\omega}$

* $\mathcal{RL}_2(j\mathbb{R})$ – real, rational, strictly proper T.M with no poles on $j\mathbb{R}$.

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$\mathcal{L}_2(j\mathbb{R})$

$\mathcal{H}_2$ $\left(\begin{array}{l} \text{analytic in } \mathbb{C}^+ \\ \text{for almost all } \omega, \lim_{\sigma \to 0^+} U(\sigma + j\omega) = U(j\omega) \\ \int_{-\infty}^{\infty} \text{trace}\left[F^*(j\omega) F(j\omega)\right] d\omega < \infty \end{array}\right)$

$\mathcal{RH}_2$ – real, rational, strictly proper T.M with no poles in $\overline{\mathbb{C}}^+$

$\mathcal{H}_2^\perp$ $\left(\begin{array}{l} \text{analytic in } \mathbb{C}^- \\ \text{for almost all } \omega, \lim_{\sigma \to 0^-} U(\sigma + j\omega) = U(j\omega) \\ \int_{-\infty}^{\infty} \text{trace}\left[F^*(j\omega) F(j\omega)\right] d\omega < \infty \end{array}\right)$

$\mathcal{RH}_2^\perp$ – real, rational, strictly proper T.M with no poles in $\overline{\mathbb{C}}^-$

Induced norm of a bounded linear operator: $\mathcal{T} : \mathcal{V} \longrightarrow \mathcal{W}$

$$\|\mathcal{T}\|_{\mathcal{V} \to \mathcal{W}} := \sup_{\substack{v \in \mathcal{V} \\ v \neq 0}} \frac{\|\mathcal{T} v\|_{\mathcal{W}}}{\|v\|_{\mathcal{V}}} = \sup\left\{ \|\mathcal{T} v\|_{\mathcal{W}} \,\middle|\, v \in \mathcal{V} \text{ and } \|v\|_{\mathcal{V}} \leq 1 \right\}$$